MATH*6210 A1

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Toy model of glycemic control:

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') dt' \tag{1}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{2}$$

1. Starting from the model with one integral and one differential equation, derive an ODE with 2 DOF

Using Leibniz's rule:

$$\dot{u}(t) = \alpha \dot{v} + \beta \left[\lambda e^{-\lambda(t-t)} v(t) \cdot (1) + \int_{-\infty}^{t} \lambda \left(-\lambda e^{-\lambda(t-t')} \right) v(t') \, dt' \right]$$

Setting $e^0 = 1$ and factoring out $-\lambda$ from the integral:

$$\dot{u} = \alpha \dot{v} + \beta\lambda v - \beta\lambda \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') \, dt'$$

$$\text{Sub in: } \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') \, dt' = u - \alpha v \quad \text{from equation (1)}$$

$$\begin{aligned} \dot{u} &= \alpha \dot{v} + \beta\lambda v - \lambda(u - \alpha v) \\ &= \alpha \dot{v} + \beta\lambda v - \lambda u + \alpha\lambda v \\ \dot{u} + \lambda u &= \alpha \dot{v} + \lambda(\alpha + \beta) v \end{aligned}$$

Substituting $\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t)$ from equation (2):

$$\dot{u} = -\lambda u + \lambda(\alpha + \beta) v + \alpha \left(-\gamma v - \delta u(v + \bar{v}) + F(t) \right) \tag{3}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{4}$$

Final form:

$$\dot{u} = -\lambda u + \left[\lambda(\alpha + \beta) - \alpha\gamma \right] v - \alpha\delta u(v + \bar{v}) + \alpha F(t) \tag{5}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{6}$$

2. Scale the dependent variables and time to reduce the number of parameters. Follow the lecture and fill the gaps.

Let the scales be defined as:

$$v = \bar{v} v', \quad u = \tilde{u} u', \quad t = \tilde{t} t'$$

Recall equations (5) and (6). Substituting into the v equation:

$$\begin{aligned} \frac{1}{\tilde{t}} \bar{v} \frac{dv'}{dt'} &= -\gamma(\bar{v} v') - \delta(\tilde{u} u')(\bar{v} v' + \bar{v}) + F \\ \implies \frac{dv'}{dt'} &= -(\gamma \tilde{t}) v' - (\delta \tilde{t} \tilde{u}) u'(v' + 1) + \frac{\tilde{t}}{\bar{v}} F \quad \text{let } G = \frac{\tilde{t}}{\bar{v}} F \end{aligned}$$

Setting coefficients to 1:

$$\gamma\tilde{t} = 1 \implies \tilde{t} = \frac{1}{\gamma}$$
$$\delta\tilde{t}\tilde{u} = 1 \implies \tilde{u} = \frac{1}{\delta\tilde{t}} = \frac{\gamma}{\delta}$$
$$\implies \underbrace{\frac{dv'}{dt'} = -v' - u'(v'+1) + G}_{\text{call this } f'(u',v') \text{ (now dimensionless)}}$$

Substituting into the u equation:

$$\frac{1}{\tilde{t}}\tilde{u}\frac{du'}{dt'} = -\lambda\tilde{u}u' + \lambda(\alpha+\beta)\bar{v}v' + \alpha f$$
$$\implies \frac{du'}{dt'} = -(\lambda\tilde{t})u' + \left[\lambda\bar{v}(\alpha+\beta)\left(\frac{\tilde{t}}{\tilde{u}}\right)\right]v' + \left(\alpha\frac{\tilde{t}}{\tilde{u}}\right)f$$

Using $\tilde{t} = \frac{1}{\gamma}$ and $\tilde{u} = \frac{\gamma}{\delta}$:

$$\implies \frac{du'}{dt'} = -\left(\frac{\lambda}{\gamma}\right)u' + \left[\left(\frac{\lambda\bar{v}}{\gamma}\cdot\frac{\delta}{\gamma}\right)(\alpha+\beta)\right]v' + \left(\alpha\frac{\delta}{\gamma^2}\right)f'(u',v')$$

Let $a = \frac{\lambda}{\gamma}$, $\quad b = \frac{\lambda\bar{v}\delta}{\gamma^2}(\alpha+\beta)$, $\quad c = \frac{\alpha\delta}{\gamma^2}$

$$\implies \frac{du'}{dt'} = -au' + bv' + cf'(u',v')$$

3. Find the equilibria if any

Drop the primes:

$$-v - u(v+1) + G = 0$$
$$-au + bv = 0 \quad \text{(since } f(u,v) = 0 \text{ from the above equation)}$$

From the second equation:

$$-au + bv = 0 \iff u = \frac{b}{a}v$$

Substituting this into the first equation:

$$-v - u(v+1) + G = 0$$
$$\iff -v - \frac{b}{a}v(v+1) + G = 0$$
$$\iff -\frac{b}{a}v^2 - \frac{b}{a}v - v + G = 0$$
$$\iff \frac{b}{a}v^2 + \left(1 + \frac{b}{a}\right)v - G = 0$$
$$\iff v^2 + v + \frac{a}{b}v - \frac{a}{b}G = 0$$
$$\iff v^2 + \left(1 + \frac{a}{b}\right)v - \frac{a}{b}G = 0$$

Using the quadratic formula to solve for the equilibrium points v^*:

$$v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4aG}{b}}$$

Analysis of Solutions

When $G > G_{\text{SN}}$, we get two separate equilibrium solutions representing our upper and lower branches—specifically, a stable node and a saddle point. If we drop down to the critical bifurcation threshold G_{SN}, the square root term vanishes and leaves us with just a single equilibrium point, defined by $G_{\text{SN}} = -\left(1 + \frac{a}{b}\right)^2 \frac{b}{4a}$.

4. Exploring Controllability via Python script

Looking at the eigenvalues for this system gives us quantitative confirmation of local stability. For the set of baseline parameters ($p_0 = 0.1, p_1 = 1.0, p_2 = 1.0$) at a parameter value of $G = 2.0$, linearization around the upper equilibrium point (eq_0) gives us a Jacobian matrix with strictly negative eigenvalues ($\lambda_1 = -0.440, \lambda_2 = -3.091$), confirming its asymptotically stable node. In contrast, at the lower equilibrium point (eq_1) consistently yields real eigenvalues of opposite signs ($\lambda_1 = -0.118, \lambda_2 = 11.549$) with a negative determinant ($\det(J) < 0$), verifying that its an unstable saddle point. In general, the existence and character of these states remain strictly governed by the sign of the system discriminant D. Looking at how the system behaves under different values of G, a pattern is showen. Higher values of G push the equilibria further apart and widen the safe zone, whereas dropping G toward the bifurcation limit shrinks the controllable region and makes the system much more prone to failure.

To visualize how the system behaves globally, the phase portrait brings everything together with vector field streamlines and nullclines. The blue ($\frac{du}{dt} = 0$) and orange ($\frac{dv}{dt} = 0$) curves cross right at our equilibrium points. By mapping out the stable manifold of the saddle point, we can clearly see the separatrix dividing the phase space. Initial states on the safe side converge to our stable node, while anything crossing that threshold spirals into runaway divergence ($v < -100$) and a total loss of glucose regulation.

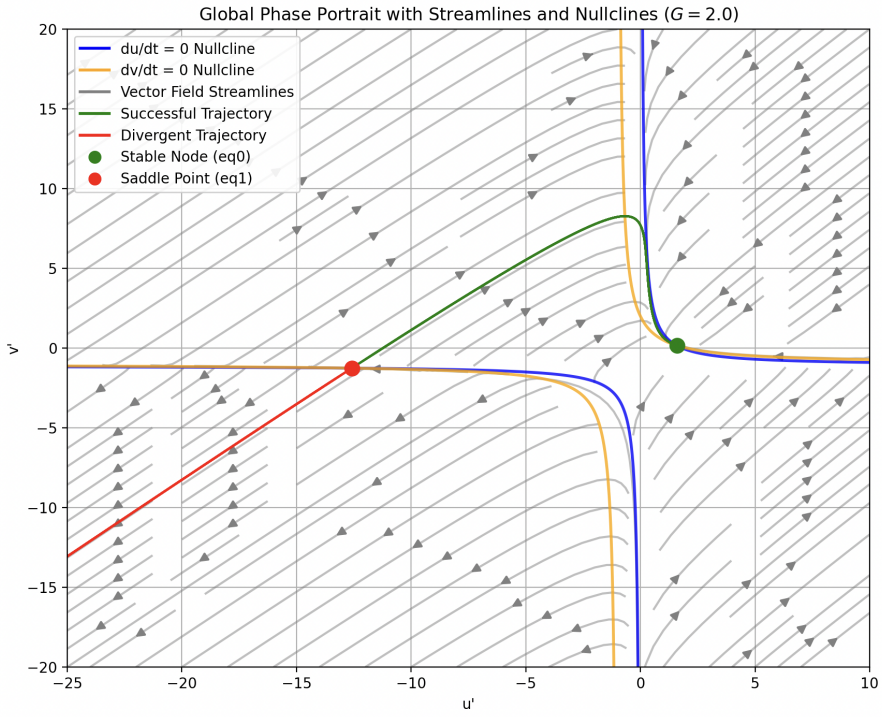


Figure 1: Global phase portrait for $G = 2.0$, featuring vector field streamlines, nullclines ($\frac{du}{dt} = 0$ in blue, $\frac{dv}{dt} = 0$ in orange), the stable node (eq_0), and the saddle point (eq_1).

Parameter changes directly dictate these boundaries. As G drops toward its critical limit G_{SN}, the stable node and saddle point migrate toward each other until they collide at the bifurcation point, where a zero eigenvalue wipes out the equilibria entirely. In the phase space, this contraction heavily shrinks our safe zone, making the system vastly more sensitive to external disturbances.